

Problem Statement

Extract actionable insights from historical data and prepare a presentation to assist the CEO in decision making

Objective 1

Which products generate the most revenue?

- What are the best selling products?
- What are the least selling products?
- What is the revenue by product category?
- What is the revenue by product type?

Objective 2

What are the sales peak times?

- What is the sales trend during the day
- What are the sales performances by each month?
- What are the sales performance by each month period?

Objective 3

Sales trends across products and time intervals?

- What are the total units sold?
- What is the sales revenue at each store location?
- What is the total revenue?

Objective 4

What are the recommendations for improving sales performance?

- Inventory adjustments
- Staffing adjustments
- Promotion opportunities for slow time intervals and slow selling products

Problem Statement
Extract actionable insights from historical data and prepare a presentation to assist the CEO in decision making

Objective 1
Which products generate the most revenue?

What are the best-selling products?
What are the top-selling products?
What are the most profitable products?
What are the most popular products?

Objective 2
What are the sales peak times?

What are the sales peaks during the day?
What are the sales peaks during the week?
What are the sales peaks during the month?

Objective 3
Sales trends across products and time intervals?

What are the sales trends across products?
What are the sales trends across time intervals?
What are the sales trends across both?

Objective 4
What are the recommendations for improving sales performance?

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Data ingestion

**Data processing/
ETL Pipeline**

Data storage

Data analysis

**Data visualisation/
Dashboard layout**

Results presentation

DATA SOURCE
BRIGHTLEARN COFFEE SHOP
SALES CSV DATA SET (FROM
LOWER MANHATTAN, HELL'S
KITCHEN AND ASTORIA
STORE LOCATIONS)

* Upload
Brightlearn
Coffee Shop
CSV dataset
into
Databricks

EXTRACT
* Read CSV file

TRANSFORM
* Handle missing values
* Remove duplicate entries
* Transform data types (unit price
into numeric using case
statement)
* Standardisation of date formats
* Feature engineering (time)

LOAD
* Store the cleaned dataset in Databricks table
for analysis

Databricks storage
* Store data in
delta tables under
the Gold layer

SQL Queries
Retrieve sales records
- Filter date ranges
- Analyse product sales

SQL Aggregations
- SUM (total_quantity x unit
price)
- SUM(transaction_qty)
- COUNT(transaction_id)
- AVG (order_value)
- GROUP BY
product_category

SQL Case statements
- Classify time periods (day
time, day of the week,
month period)
- Classify expenditure
patterns (low, medium and
high spend)

KPI
- Total revenue
- Total units sold
- Total transactions

METRICS
- Revenue by store location
- Transaction quantity by
product category
- Revenue by product type
- Top performing products
- Least performing products
- Product sales by time of day
- Product sales by month
period
- Product sales by month

KPI CARDS
TOTAL
UNITS
SOLD
REVENUE
BY STORE
LOCATIONS
TOTAL
REVENUE

TREND ANALYSIS
REVENUE TREND
(LINE GRAPH)
SALES BY TIME
INTERVALS (LINE
GRAPH)

PRODUCT PERFORMANCE
REVENUE BY STORE
LOCATION (DONUT
CHART)
TRANSACTION
QUANTITY BY
PRODUCT CATEGORY
(COLUMN GRAPH)
REVENUE BY
PRODUCT TYPE
(COLUMN GRAPH)
TOP AND LEAST
PERFORMING
PRODUCTS (BAR
GRAPH)

INSIGHTS PANEL
- Best selling products
- Highest revenue category
- Peak sales time
- Best performing stores

EXECUTIVE SUMMARY
Give a summary of the objectives, key
findings, limitations and
recommendations.

ANALYSIS OBJECTIVES
Sales performance by
- Store location
- Product category
- Product type
- Time periods
- Recommendations

KEY BUSINESS INSIGHTS
- Total revenue
- Total transactions
- Revenue by store location
- Best and least performing products
- Peak and off-peak times

RECOMMENDATIONS
- Stocking
- Staffing
- Sales promotions